Yan DAI (戴言)

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Education

Massachusetts Institute of Technology Operations Research Center — Sep 2024 – May 2028 (Expected)
PhD student in Operations Research co-advised by Patrick Jaillet and Negin Golrezaei.

Tsinghua University Andrew C. Yao Class — Sep 2020 – Jun 2024
Bachelor of Engineering in Computer Science and Technology. GPA: **3.99/4.00**.

Research Interests

EconCS: mechanism design, pricing, and resource allocation with feedback from strategic agents.
Learning Theory: bandits, online learning, learning in games, and reinforcement learning theory.
Broader Interests: information economics, queueing networks, and optimization for deep learning.

Publications and Working Papers

Mechanism Design, Pricing, and Resource Allocation

[17] **[NeurIPS'25]** Yan Dai, Negin Golrezaei, Patrick Jaillet. "Efficiency, Feasibility, and Incentive-Awareness in Constrained Online Resource Allocation." **Under Review** at *Operations Research*.
Early version won **1st Place** in ACM Student Research Competition (SRC), SIGMETRICS'25.

[16] **[COLT'25]** Yan Dai, Moïse Blanchard, Patrick Jaillet. "Non-Monetary Mechanism Design without Priors: Achieving Efficiency via Adaptive Costly Audits." **Major Revision** at *Operations Research*.

[15] [*Under Review*] Yan Dai, Tao Lin, Aleksandrs Slivkins. "The Power of Two Choices in Dynamic Pricing: Near-Instance-Optimal Regret and Query Complexity."

[14] [*Working Paper*] Yan Dai, Nicole Immorlica, Brendan Lucier, Aleksandrs Slivkins. "The Speed of Social Learning with Pricing."

[13] [*Working Paper*] Yan Dai, Maryam Farboodi, Negin Golrezaei, Sepehr Shahshahani. "Market Design for AI: Beyond the Copyright Binary." Early version in 2026 M&SOM Service Operations SIG (acceptance: 10/96).

Bandits and Online Learning Theory

[12] **[SIGMETRICS'25 Best Paper Award]** Yan Dai and Longbo Huang. "Adversarial Network Optimization under Bandit Feedback: Maximizing Utility in Non-Stationary Multi-Hop Networks."

[11] **[ICLR'25 Spotlight (5%)]** Yu Chen*, Jiatai Huang*, Yan Dai*, Longbo Huang. "uniINF: Best-of-Both-Worlds Algorithm for Parameter-Free Heavy-Tailed MABs."

[10] **[ICML'23]** Jiatai Huang*, Yan Dai*, Longbo Huang. "Banker Online Mirror Descent: A Universal Approach for Delayed Online Bandit Learning."

[9] **[ICLR'23]** Yan Dai, Ruosong Wang, Simon S. Du. "Variance-Aware Sparse Linear Bandits."

[8] **[ICML'22]** Jiatai Huang*, Yan Dai*, Longbo Huang. "Adaptive Best-of-Both-Worlds Algorithm for Heavy-Tailed Multi-Armed Bandits."

[7] [*Under Review*] Yan Dai, Negin Golrezaei, Patrick Jaillet. "Policy Regret for Embedding Model Routing: Contextual Bandits with Low-Rank Experts."

Reinforcement Learning Theory

[6] [*Under Review*] Aarush Kulkarni, Khang Nguyen, William Chang, Yan Dai. "Introducing Feedback Delays to Non-Delayed MDPs: Optimal $\sqrt{S}$ and $\sqrt{K}$ Dependencies in Tabular and Linear AMDPs." *Mentoring Project.*

[5] **[COLT'24]** Yan Dai, Qiwen Cui, Simon S. Du. "Refined Sample Complexity for Markov Games with Independent Linear Function Approximation."

[4] **[ICML'23]** Yan Dai, Haipeng Luo, Chen-Yu Wei, Julian Zimmert. "Refined Regret for Adversarial MDPs with Linear Function Approximation."

[3] **[NeurIPS'22]** Yan Dai, Haipeng Luo, Liyu Chen. "Follow-the-Perturbed-Leader for Adversarial Markov Decision Processes with Bandit Feedback."

Optimization for Deep Learning

[2] **[ICML'24]** Kwangjun Ahn, Zhiyu Zhang, Yunbum Kook, Yan Dai. "Understanding Adam Optimizer via Online Learning of Updates: Adam is FTRL in Disguise."

[1] **[NeurIPS'23]** Yan Dai*, Kwangjun Ahn*, Suvrit Sra. "The Crucial Role of Normalization in Sharpness-Aware Minimization."

Selected Awards

- Runner-Up of Two Sigma PhD Fellowship *(2 winners and 2 runners-up)*. Feb 2026
- **Best Paper Award** *(1 paper per year)* of ACM SIGMETRICS 2025. Jun 2025
- Outstanding Graduate named by Beijing *(5%)*, by Tsinghua *(2%)*, and by Yao Class *(10%)*. Jun 2024
- **Presidential Scholarship** *(top scholarship for Tsinghua undergrads; 10/16,000+)*, Tsinghua. Nov 2023
- Andrew C. Yao Award – Gold Medal *(top scholarship for seniors; 1 student)*, Yao Class. Sep 2023
- Gold Medal *(1st place with perfect score)*, Asia-Pacific Informatics Olympiad (APIO). May 2019
- Gold Medal *(5th place)*, Chinese National Olympiad in Informatics (NOI). Jul 2018

Research Experience

Massachusetts Institute of Technology OR Center (Cambridge, MA) 2024 – Now

Research assistant of Patrick Jaillet and Negin Golrezaei. [17], [16], [13], [7].

Microsoft Research EconCS Group (Cambridge, MA) May – Aug 2026

Research intern of Alex Slivkins and Brendan Lucier (also worked with Nicole Immorlica). [15], [14].

Tsinghua University Andrew C. Yao Class (Beijing, China) Feb – Jun 2024

Bachelor's thesis advised by Longbo Huang. [12] (Best Paper Award of SIGMETRICS'25).

University of Washington School of CSE (Seattle, WA) Jun – Aug 2023

Visiting student hosted by Simon S. Du. [9], [5].

Massachusetts Institute of Technology Lab for Info. & Decision (Cambridge, MA) Feb – May 2023

Visiting student hosted by Suvrit Sra. [2], [1].

University of Southern California CS Department (Remote) Feb – Dec 2022

Remote visitor of Haipeng Luo. [4], [3].

Tsinghua University Andrew C. Yao Class (Beijing, China) Jun – Dec 2021

Research assistant of Longbo Huang. [11] (Spotlight of ICLR'25), [10], [8].

Professional Service

- **Operations Research & Economics:** *Operations Research* (OR); *Mathematics of Operations Research* (MOR); *Games and Economic Behavior* (GEB).
- **Machine Learning:** *Journal of Machine Learning Research* (JMLR), *IEEE Transactions on Pattern Analysis and Machine Intelligence* (TPAMI), *Machine Learning* (MLJ), *Transactions on Machine Learning Research* (TMLR).
- **Networking & Systems:** *IEEE/ACM Transactions on Networking* (TON), *Performance Evaluation* (PEVA), *IEEE Internet of Things Journal* (IoT-J).
- **Conferences:** COLT 2025; ICML 2026 (gold reviewer) / 2025 / 2023; NeurIPS 2026 / 2025 / 2024 / 2023 / 2022; ICLR 2027 / 2025 / 2024; AISTATS 2027 / 2026 / 2025 / 2023 / 2022; AAAI 2027 / 2026; ALT 2023.